

MONASH University

Are current sortition methods creating representative panels? A framework analysis

Daniel Loza
B.Bus., Monash University

A thesis submitted for the degree of
Bachelor of Commerce (Honours)
at Monash University in 2026
Department of Econometrics & Business Statistics

Table of contents

Copyright notice iv

Abstract v

Declaration vi

Acknowledgements viii

1 Introduction 1

2 Literature Review 3
- 2.1 Representation and Sortition . . . 3
- 2.2 Panel Selection . . . 4
- 2.3 Visualisation Methods . . . 6

3 Methodology 8
- 3.1 Research Question . . . 8
- 3.2 Research Design . . . 8
- 3.3 Research Plan . . . 8

4 Preliminary Results 9

Bibliography 10

Copyright notice

Produced on 16 April 2026.

© Daniel Loza (2026).

Abstract

Sortition, the random selection of individuals to form panels, has seen a sort of renaissance in recent years, as more and more institutions realise the value of direct community and stakeholder input when making decisions. These panels have proven to be massively beneficial in terms of improving the quality of decisions, increasing public trust and engagement, and creating more inclusive and representative decision-making processes. However, there are still many challenges to overcome in order to ensure that these panels are truly representative and effective. This thesis aims to evaluate current sortition methods and community representation processes, as well as to investigate how unobserved traits and survey non-response may affect panel representativeness. By investigating these factors, we can develop a greater understanding of existing tools and procedures, how they work, how they fall short, and what it would mean for them to be more representative, which will then enable the creation of a diagnostic tool to gauge effectiveness.

Declaration

Use only one of the following declarations (Standard thesis or Thesis including published works declaration) and remove the other.

Standard thesis

This thesis is an original work of my research and contains no material which has been accepted for the award of any other degree or diploma at any university or equivalent institution and that, to the best of my knowledge and belief, this thesis contains no material previously published or written by another person, except where due reference is made in the text of the thesis.

Student name: Daniel Leo Loza

Student signature:

Date:

Publications during enrolment

Remove this section if you do not have publications.

The material in Chapter 1 has been submitted to the journal *Journal of Impossible Results* for possible publication.

The contribution in Chapter 2 of this thesis was presented in the International Symposium on Nonsense held in Dublin, Ireland, in July 2022.

Reproducibility statement

This thesis is written using Quarto with renv (**renv**) to create a reproducible environment. All materials (including the data sets and source files) required to reproduce this document can be found at the Github repository `github.com/SusanSu/thesis`.

This work is licensed under a Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License.

Thesis including published works declaration

I hereby declare that this thesis contains no material which has been accepted for the award of any other degree or diploma at any university or equivalent institution and that, to the best of my knowledge and belief, this thesis contains no material previously published or written by another person, except where due reference is made in the text of the thesis.

This thesis includes ?? original papers published in peer reviewed journals and ?? submitted publications. The core theme of the thesis is ??. The ideas, development and writing up of all the papers in the thesis were the principal responsibility of myself, the student, working within the Department of Econometrics & Business Statistics under the supervision of ??

(The inclusion of co-authors reflects the fact that the work came from active collaboration between researchers and acknowledges input into team-based research.)

In the case of (??insert chapter numbers) my contribution to the work involved the following:

Thesis chapter	Publication title	Status	Nature and % of student contribution	Nature and % of coauthors' contribution	Coauthors are Monash students
2	The life cycle of Mongolian crickets	Submitted	Concept and data analysis, writing first draft: 60%	Shu Xu, input into manuscript: 25%; Eddie Betts, input into manuscript: 15%	Shu Xu: No; Eddie Betts: Yes

I have / have not renumbered sections of submitted or published papers in order to generate a consistent presentation within the thesis.

Student name:

Student signature:

Date:

I hereby certify that the above declaration correctly reflects the nature and extent of the student's and co-authors' contributions to this work. In instances where I am not the responsible author I have consulted with the responsible author to agree on the respective contributions of the authors.

Main Supervisor name:

Main Supervisor signature:

Date:

Acknowledgements

I would like to thank my pet goldfish for ...

In accordance with Chapter 7.1.4 of the research degrees handbook, if you have engaged the services of a professional editor, you must provide their name and a brief description of the service rendered. If the professional editor's current or former area of academic specialisation is similar your own, this too should be stated as it may suggest to examiners that the editor's advice to the student has extended beyond guidance on English expression to affect the substance and structure of the thesis.

If you have used generative artificial intelligence (AI) technologies, you must include a written acknowledgment of the use and its extent. Your acknowledgement should at a minimum specify which technology was used, include explicit description on how the information was generated, and explain how the output was used in your work. Below is a suggested format:

"I acknowledge the use of [insert AI system(s) and link] to [specific use of generative artificial intelligence]. The output from these was used to [explain use]."

Free text section for you to record your acknowledgment and gratitude for the more general academic input and support such as financial support from grants and scholarships and the non-academic support you have received during the course of your enrolment. If you are a recipient of the "Australian Government Research Training Program Scholarship", you are required to include the following statement:

> "This research was supported by an Australian Government Research Training Program (RTP) Scholarship."

You may also wish to acknowledge significant and substantial contribution made by others to the research, work and writing represented and/or reported in the thesis. These could include significant contributions to: the conception and design of the project; non-routine

technical work; analysis and interpretation of research data; drafting significant parts of the work or critically revising it to contribute to the interpretation.

Chapter 1

Introduction

(got advised in class to put aim first, but then was also told it depends on context; will need to discuss this again!)

This thesis aims to evaluate whether current sortition methods and ideologies lead to actually representative panels in practice. Through simulation and visualisation of algorithm performance, we will determine whether current tools are performing as expected and producing panels that are actually representative.

Sortition is a method of selection that uses random sampling to create panels that should be descriptively representative of a population. It has been used in various contexts, reaching as far back as ancient Athens, where it was how jurors were selected, and into modern times for citizens' assemblies. The idea is that by randomly selecting individuals from the population, we can create panels that are more representative and inclusive than those selected through traditional methods. Sortition also reduces dominance by elites or highly engaged groups, enabling broader participation across social classes (Giuliani 2025), making it more likely that decisions are in the relevant population's best interest.

According to Flanigan et al. (2021), sortition creates groups that are statistical "mini-publics" of society, reflecting demographics and perspectives more accurately than elected bodies. These sorts of panels enable more inclusive and representative decision-making processes, which can lead to better outcomes and increased public trust and engagement. They also encourage more diverse perspectives and can help to break down barriers between different groups in society, fostering greater understanding and collaboration.

An increasing number of institutions, including the Parisian (*Paris Creates a Permanent Citizens' Council* 2026), Irish (Walsh & Elkink 2021), and British Columbian (Ward 2008) governments, are using

sortition to create panels that are more representative of the population for their political discussions and deliberations, so the need for robust and fair methods for building representative panels is only growing. However, there are still many challenges to overcome in order to ensure that these panels are truly representative and effective for the populations they aim to serve.

One of the main roadblocks is ensuring that the selection process is transparent and fair, and that the panels are truly representative of the population in terms of demographics and perspectives. But this raises the questions of what it means to be fair, what it means to be representative, and how these look from a systematic perspective.

Additionally, though some tools (such as LexiMin, from Flanigan et al. (2021)) have already been developed, there is a need for better methods and frameworks to evaluate the representativeness of these panels and to ensure that they are robust in achieving their goals.

This study aims to evaluate current sortition methods and community representation processes, as well as to investigate how unobserved traits and survey non-response may affect panel representativeness. By investigating these factors, we can develop a greater understanding of existing tools and procedures, how they work, how they fall short, what it would mean for them to be more representative, which will then enable the creation of a diagnostic tool to gauge effectiveness.

The remainder of this thesis is structured as follows: Chapter 2 provides a literature review of the existing research on sortition and representation, and identifies gaps in the current knowledge. Chapter 3 outlines the methodology used to evaluate the representativeness of panels and to investigate the impact of unobserved traits and survey non-response. Chapter 4 presents the results of the analysis, and Chapter 5 discusses the implications of the findings and suggests directions for future research.

Chapter 2

Literature Review

2.1 Representation and Sortition

2.1.1 What does it mean to be representative?

The main aim of any and all selection methods within sortition is to create a panel that is fair and representative of the population. But what does this actually mean?

Fairness refers to the idea that the selection process should be transparent, unbiased, and inclusive. It should give everyone an equal chance of being selected, regardless of their background or characteristic, which presents a likelihood difference minimisation challenge. Systematically, this is typically achieved through introducing randomness to selection processes (Flanigan et al. 2021), as not doing so immediately introduces implicit selection bias.

Alongside this, representativeness refers to the idea that the selected panel should accurately reflect the demographics and perspectives of the given population. This means that the panel should be diverse and inclusive, and should contain individuals from a wide range of relevant backgrounds. This can be assessed directly by comparing demographic compositions of the selected panel against the population from which it was originally drawn.

Approaching it systematically, the sortition process can be thought of in four stages:

- Defining the population and traits of interest
- Sampling from the population (through survey or other outreach)
- Panel selection (through algorithmic means or other method)
- Panel evaluation, confirmation and engagement

Each of these steps presents an opportunity for bias to be introduced, and for the panel to become less representative of the population. Which is why it is crucial to evaluate the level of representativeness (or *potential* representativeness) at each stage, and to understand how different factors may affect it.

For example, if the population of interest and the investigated traits are poorly defined, then the selection process may not be able to capture the diversity and perspectives of the desired group, leading to a less representative panel. This presents a challenge in itself, given most population descriptions (as previously discussed) are often based on disjoint demographic characteristics and do not provide a way to capture intersectionality, which is a key aspect of representativeness. However, this may be beyond the scope of this thesis, as it is already a foray into broader data collection and storage methods, but it is important to keep in mind that representation goes beyond the margin.

Similarly, if certain groups are less likely to respond to the survey, then they may be underrepresented in the pool of potential panelists, which can lead to a less representative panel from the outset. Illustratively–if there were to be no female respondents for a certain survey in an area comprised of 60% women, then the representativeness assumption (and goal) would be majorly violated before an attempt at putting a panel together has even been made. Situations like these raise the question of how we can account for this non-response bias, and whether we can use visualisation tools to show the discrepancy between the pool and the population (which leads back to part of this thesis' aim).

2.2 Panel Selection

2.2.1 Problem Definition

Suppose there is a pool of individuals numbered 1 to n.

We want to choose a panel of size k.

Let K be the set of all feasible panels. A feasible panel is any group of k individuals that satisfies the required constraints (which generally take the form of quotas).

These quotas will be certain desired counts, or ranges of counts, of individuals belonging to a certain trait class (e.g. 20-22 males, or 15-17 rural). These traits and quotas, by extension, will have direct ties to the demographics that have been deemed crucial to represent for a certain project's population group. Tying back to the established background–the core goal of sortition, then, is to build a panel which meets the quota constraints, equitably, robustly, and in a way that represents the population from which it was drawn.

2.2.2 Existing Methods

Non-algorithmic/self-selection

This is the most basic method of panel selection, where individuals are simply allowed to self-select into the panel. This can be done through a variety of means, such as advertising the opportunity to participate in the panel and allowing individuals to sign up on a first-come, first-served basis. However, these types of methods are often not very effective at creating representative panels, as they can a) lead to self-selection bias (as involvement will be tied to exposure and not random chance, those who put their hand up of their own volition are more likely to have pre-formed perspectives, ahead of the panel process), thus possibly not capturing the diversity and perspectives of the population, and b) lead to lower-than-desired engagement, as it does not guarantee any particular level of exposure or direct engagement in the first place.

LEGACY heuristic (I can't find original source, but it is mentioned in Flanigan et al. (2021))

Already a systematic improvement over non-algorithmic methods; the LEGACY selection heuristic takes a given sample of respondents and selects individuals one-by-one until the panel is full. At each step, the heuristic calculates a "need" value to identify the trait that is currently furthest from the desired count in the panel. LEGACY then selects a random individual from the pool who possesses that trait. If selecting any given individual would cause the panel to hit the upper quota for any trait, then all other individuals with that trait are removed from the pool before the next selection. This process is repeated until it either becomes infeasible (exhausts the dataset before hitting overall quota, at which point it will restart), or builds a full panel with desired quota representation. LEGACY's process fulfils the base requirements of building a panel that meets the desired quotas, but it does not necessarily maximise representativeness, as it does not take into account the likelihood of selection for different individuals (allowing them to just be 0%, in extreme cases), and it does not attempt to minimise the maximum deviation from equal selection chance across all individuals.

MaxiMin and MiniMax (Baharav & Flanigan 2024)

This aforementioned shortcoming is then addressed by our first algorithmic sorition methods: MaxiMin and MiniMax, though these both tackle the problem in opposite ways and with differing goals. MaxiMin, as the name would suggest, is a selection algorithm that generates potential panels and selection probabilities and then sets out to maximise the minimum selection probability of the set. This reduces the likelihood of an individual never being chosen (improves fairness) but makes the method quite susceptible to manipulation, if an individual elects as belonging to certain quota groups with lesser representation in the sample. Thus, leading to potential representative burden. Inversely, MiniMax aims to minimise the maximum probability of an individual being selected, with the aim of making selection more robust (less vulnerable to manipulation). However, this then comes at the

cost of fairness, as the issue of potential 0% selection likelihood returns.

LexiMin (Flanigan et al. 2021)

LexiMin, or lexicographic minimisation, is a method of panel selection similar to MaxiMin, that aims to maximise the minimum likelihood of selection, but for each individual being selected sequentially (rather than just the outright minimum). It works by first identifying the individual with the lowest selection chance, and then selecting panel combinations that make this individual's selection chance as high as possible. This process is repeated for each individual in order of their selection likelihood until overall selection chance is maximised and, ideally, made as equal as possible across all participants. This ensures that each individual has the highest possible chance of being selected. But this trait of LexiMin can, like MaxiMin, make it vulnerable to manipulation.

Goldilocks (Baharav & Flanigan 2024)

Bringing us to the supposed current gold standard: the aptly named "Goldilocks" sortition algorithm. Goldilocks is structured in such a way that it simultaneously attempts to maximise minimum selection probabilities (improve fairness) and minimise maximum selection probabilities (improve robustness), providing a "best of both worlds" solution to the problem by averaging outcomes.

2.2.3 New Investigation

Across all of these previous algorithmic techniques, it is noted that stratification for combinations of demographic traits is not included within their decision-making designs–focusing instead on each trait at the margin. This is a major limitation, as it means that the selection process may not be able to entirely capture the diversity and perspectives of the desired group, leading to a less representative panel. In some situations (e.g. supporting side-panels built for specific subgroups, communities where there are known intersectional minorities of interest), it may be necessary to ensure the inclusion of specific intersectional combinations of traits in order to truly represent a given population.

The extent to which this happens naturally through current methods will be evaluated in the later simulation. We will then also test the effects of further imposing combinational quotas on the representative burden, and on the selection probabilities of different individuals, to see whether this is a feasible way to ensure the representation of intersectional minorities in situations where it is deemed necessary.

2.3 Visualisation Methods

In order to evaluate the representativeness of selected panels, we will use a variety of visualisation methods to show how the distribution of traits in the selected panel compares to the overall popu-

lation, and to identify any discrepancies or biases in the selection process. Some potentially useful visualisation methods include: bar charts, mosaic plots, double decker plots, UpSet plots and parallel coordinate plots.

First, it will be of use to just do some illustrative graphing to show densities within the distribution across traits, in the form of bar charts. These will give a rough visual idea of the breakdown of the population and samples within the simulation, setting up some expectations for how representation should look in further visualisations. From there, either mosaic or double decker plots will be useful for depicting the proportion of given demographic traits that are selected for the panels. Finally, UpSet plots (Conway, Lex & Gehlenborg 2017) will be an effective way to showcase the intersections that exist in generated panels, as they offer a visual link between demographic variables, as well as frequency counts to highlight how often these occur within a given panel.

– ignore the summaries –

2.3.1 Parallel coordinate plots (VanderPlas et al. 2023)

- Can help to show the distribution of traits across the selected panel and how it compares to the overall population.
- Can also show the trade-offs between different traits and how they are balanced in the selection process
- Can be used to identify any segments of the population that are underrepresented in the selected panel, and to highlight any potential biases in the selection process.

2.3.2 Matrix-based set representation (UpSetR) (Conway, Lex & Gehlenborg 2017)

- Can be used to show how different traits connect with each other and how they are distributed across the distribution and/or selected panel.
- Can help to identify any clusters of traits that are overrepresented or underrepresented in the selected panel, and to show how different traits interact with each other in the selection process.
- Can highlight specific combinations of traits that are of interest, and show how they are represented in the selected panel compared to the overall population. (Though there is no way to gauge the prevalence of these combinations in the population, beyond anecdotal observations)

–

2.3.3 Attribute distribution analysis

- Showcase how the distribution of traits in the selected panel compares to the overall population, and to identify any discrepancies or biases in the selection process.

Chapter 3

Methodology

3.1 Research Question

What does it mean for a panel to be “good” at being representative and are we achieving this? - Are current methods and algorithms, like LexiMin, performing the best they could be? - How do unobserved/unfiltered traits affect response rates, and therefore panel composition and representativeness? - How can survey non-response and other sampling-level issues bias representativeness?

nDF suggests 10x response to panel count; not generally feasibly, more often get 3x. How much of a difference does this make to burden and selection likelihood?

3.2 Research Design

3.3 Research Plan

Plan

Chapter 4

Preliminary Results

This chapter contains a summary of the context in which your research is set.

Bibliography

Baharav, C & B Flanigan (2024). Fair, Manipulation-Robust, and Transparent Sortition. *Proceedings of the 25th ACM Conference on Economics and Computation*, 756–775. (Visited on 04/15/2026).

Conway, JR, A Lex & N Gehlenborg (2017). UpSetR: An R Package for the Visualization of Intersecting Sets and Their Properties. *Bioinformatics* **33**(18). Ed. by J Hancock, 2938–2940. (Visited on 04/09/2026).

Flanigan, B, P Gölz, A Gupta, B Hennig & AD Procaccia (2021). Fair algorithms for selecting citizens' assemblies. *Nature* **596**, 548–552.

Giuliani, A (2025). Sortition in Politics: From History to Contemporary Democracy. *Parliaments, Estates and Representation* **45**(3), 317–335.

Paris Creates a Permanent Citizens' Council (2026). https://www.sortitionfoundation.org/paris_creates_permanent_citi (Visited on 04/14/2026).

VanderPlas, S, Y Ge, A Unwin & H Hofmann (2023). Penguins Go Parallel: A Grammar of Graphics Framework for Generalized Parallel Coordinate Plots. *Journal of Computational and Graphical Statistics* **32**(4), 1572–1587. (Visited on 04/09/2026).

Walsh, CD & JA Elkink (2021). The Dissatisfied and the Engaged: Citizen Support for Citizens' Assemblies and Their Willingness to Participate. *Irish Political Studies* **36**(4), 647–666. (Visited on 04/15/2026).

Ward, I (2008). An Experiment in Political Communication: The British Columbia Citizens' Assembly on Electoral Reform. *Australian Journal of Political Science* **43**(2), 301–315. (Visited on 04/15/2026).